

ALFAAH37870

## 2,4-Difluorobenzylmagnesium bromide, 0.25M in 2-Methyltetrahydrofuran

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:** 2,4-二氟苄基异丙溴化镁, 0.25M 2-甲基四氢呋喃溶液  
**Product Description:** 2,4-Difluorobenzylmagnesium bromide, 0.25M in 2-Methyltetrahydrofuran

**Cat No. :** H37870  
**Molecular Formula** C7 H5 BrF2 Mg

**Supplier** Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
Shore Road, Heysham  
Lancashire, LA3 2XY,  
United Kingdom  
Office Tel: +44 (0) 1524 850506  
Office Fax: +44 (0) 1524 850608

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
Dark grey

**Odor**  
No information available

#### Emergency Overview

Highly flammable liquid and vapor. May be harmful in contact with skin. Harmful if swallowed. Causes severe skin burns and eye damage. May form explosive peroxides. Moisture sensitive. Air sensitive.

#### Classification of the substance or mixture

|                                   |              |
|-----------------------------------|--------------|
| Flammable liquids.                | Category 2   |
| Acute Oral Toxicity               | Category 4   |
| Acute Dermal Toxicity             | Category 5   |
| Skin Corrosion/Irritation         | Category 1 B |
| Serious Eye Damage/Eye Irritation | Category 1   |

#### Label Elements



## 2,4-Difluorobenzylmagnesium bromide, 0.25M in 2-Methyltetrahydrofuran

## Signal Word

## Danger

## Hazard Statements

H225 - Highly flammable liquid and vapor

H302 - Harmful if swallowed

H313 - May be harmful in contact with skin

H314 - Causes severe skin burns and eye damage

## Precautionary Statements

## Prevention

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P271 - Use only outdoors or in a well-ventilated area

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P240 - Ground and bond container and receiving equipment

P242 - Use non-sparking tools

P243 - Take action to prevent static discharges

P280 - Wear protective gloves/protective clothing/eye protection/face protection

## Response

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P330 - Rinse mouth

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

P331 - Do NOT induce vomiting

P362 + P364 - Take off contaminated clothing and wash it before reuse

## Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Physical and Chemical Hazards

Vapors may cause flash fire or explosion. Highly flammable. May form explosive peroxides.

## Health Hazards

Harmful if swallowed. May be harmful in contact with skin. Corrosive. Causes skin and eye burns.

## Environmental hazards

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants.

This product does not contain any known or suspected endocrine disruptors.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component                           | CAS No      | Weight % |
|-------------------------------------|-------------|----------|
| Methyltetrahydrofuran               | 96-47-9     | 88.4     |
| 2,4-Difluorobenzylmagnesium bromide | 546122-71-8 | 11.6     |

## SECTION 4. FIRST AID MEASURES

## General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

## Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

## Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.

**2,4-Difluorobenzylmagnesium bromide, 0.25M in 2-Methyltetrahydrofuran****Inhalation**

If not breathing, give artificial respiration. Remove from exposure, lie down. Call a physician immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

**Ingestion**

Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.

**Most important symptoms and effects**

Causes burns by all exposure routes. Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Dry sand. Carbon dioxide (CO<sub>2</sub>). Powder. Do not use water or foam. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

## 2,4-Difluorobenzylmagnesium bromide, 0.25M in 2-Methyltetrahydrofuran

**Storage**

Corrosives area.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment****Eye Protection**

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection**

Protective gloves

| Glove material    | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|-------------------|-----------------------------------|-----------------|-------------|-----------------------|
| Disposable gloves | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use**

In case of insufficient ventilation, wear suitable respiratory equipment

**Small scale/Laboratory use**

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

No information available.

**SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES****Appearance**

Dark grey

**Physical State**

Liquid

**Odor**

No information available

**Odor Threshold**

No data available

**pH**

No information available

## 2,4-Difluorobenzylmagnesium bromide, 0.25M in 2-Methyltetrahydrofuran

|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Melting Point/Range                     | No data available        |                                   |
| Softening Point                         | No data available        |                                   |
| Boiling Point/Range                     | No information available |                                   |
| Flash Point                             | No information available | Method - No information available |
| Evaporation Rate                        | No data available        |                                   |
| Flammability (solid,gas)                | No information available |                                   |
| Explosion Limits                        | No data available        |                                   |
| Vapor Pressure                          | 23 hPa @ 20 °C           |                                   |
| Vapor Density                           | No data available        | (Air = 1.0)                       |
| Specific Gravity / Density              | No data available        |                                   |
| Bulk Density                            | No data available        |                                   |
| Water Solubility                        | Immiscible               |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | No data available        |                                   |
| Viscosity                               | No data available        |                                   |
| Explosive Properties                    | No information available |                                   |
| Oxidizing Properties                    | No information available |                                   |
| Molecular Formula                       | C7 H5 BrF2 Mg            |                                   |
| Molecular Weight                        | 231.32                   |                                   |

## SECTION 10. STABILITY AND REACTIVITY

|                          |                                    |
|--------------------------|------------------------------------|
| Stability                | Air sensitive. Moisture sensitive. |
| Hazardous Reactions      | None under normal processing.      |
| Hazardous Polymerization | No information available.          |
| Conditions to Avoid      | None known.                        |
| Materials to avoid       | Strong bases. Oxidizing agent.     |

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen fluoride. Hydrogen bromide. Metal oxides.

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

(a) acute toxicity;  
Toxicology data for the components

| Component             | LD50 Oral              | LD50 Dermal           | LC50 Inhalation      |
|-----------------------|------------------------|-----------------------|----------------------|
| Methyltetrahydrofuran | 300-2000 mg/kg ( Rat ) | 4500 mg/kg ( Rabbit ) | 6000 ppm ( Rat ) 4 h |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

## 2,4-Difluorobenzylmagnesium bromide, 0.25M in 2-Methyltetrahydrofuran

|                                            |                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (e) germ cell mutagenicity;                | No data available                                                                                                                                                                                                                                                                                                                                                               |
| (f) carcinogenicity;                       | No data available<br>There are no known carcinogenic chemicals in this product                                                                                                                                                                                                                                                                                                  |
| (g) reproductive toxicity;                 | No data available                                                                                                                                                                                                                                                                                                                                                               |
| (h) STOT-single exposure;                  | No data available                                                                                                                                                                                                                                                                                                                                                               |
| (i) STOT-repeated exposure;                | No data available                                                                                                                                                                                                                                                                                                                                                               |
| Target Organs                              | No information available.                                                                                                                                                                                                                                                                                                                                                       |
| (j) aspiration hazard;                     | No data available                                                                                                                                                                                                                                                                                                                                                               |
| Symptoms / effects, both acute and delayed | Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |

## SECTION 12. ECOLOGICAL INFORMATION

|                     |                                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Ecotoxicity effects | Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants. |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------|

| Component             | Freshwater Fish                                                  | Water Flea                                             | Freshwater Algae                                   | Microtox |
|-----------------------|------------------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------|----------|
| Methyltetrahydrofuran | LC50 (96h) > 100 mg/l<br>Onchorhynchus mykiss<br>(Rainbow trout) | Chronic NOEC >=120<br>mg/l (21 days, Daphnia<br>magna) | NOEC >= 104 mg/l<br>(72h)<br>EC50 > 104 mg/l (72h) |          |

|                               |                          |
|-------------------------------|--------------------------|
| Persistence and Degradability | No information available |
|-------------------------------|--------------------------|

| Component                                 | Degradability |
|-------------------------------------------|---------------|
| Methyltetrahydrofuran<br>96-47-9 ( 88.4 ) | (2%) 28 days  |

|                           |                          |
|---------------------------|--------------------------|
| Bioaccumulative Potential | No information available |
|---------------------------|--------------------------|

|                  |                          |
|------------------|--------------------------|
| Mobility in soil | No information available |
|------------------|--------------------------|

|                                 |                                                                           |
|---------------------------------|---------------------------------------------------------------------------|
| Endocrine Disruptor Information | This product does not contain any known or suspected endocrine disruptors |
| Persistent Organic Pollutant    | This product does not contain any known or suspected substance            |
| Ozone Depletion Potential       | This product does not contain any known or suspected substance            |

## SECTION 13. DISPOSAL CONSIDERATIONS

|                                     |                                                                                                                                                                                                                             |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waste from Residues/Unused Products | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
| Contaminated Packaging              | Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.                                                                                                                  |

## 2,4-Difluorobenzylmagnesium bromide, 0.25M in 2-Methyltetrahydrofuran

## Other Information

Waste codes should be assigned by the user based on the application for which the product was used.

## SECTION 14. TRANSPORT INFORMATION

## Road and Rail Transport

UN-No UN2924  
Proper Shipping Name Flammable liquid, corrosive, n.o.s.  
Technical Shipping Name (METHYLTETRAHYDROFURAN, 2,4-Difluorobenzylmagnesium bromide)  
Hazard Class 3  
Subsidiary Hazard Class 8  
Packing Group II

## IMDG/IMO

UN-No UN2924  
Proper Shipping Name Flammable liquid, corrosive, n.o.s.  
Technical Shipping Name (METHYLTETRAHYDROFURAN, 2,4-Difluorobenzylmagnesium bromide)  
Hazard Class 3  
Subsidiary Hazard Class 8  
Packing Group II

## IATA

UN-No UN2924  
Proper Shipping Name Flammable liquid, corrosive, n.o.s.  
Technical Shipping Name (METHYLTETRAHYDROFURAN, 2,4-Difluorobenzylmagnesium bromide)  
Hazard Class 3  
Subsidiary Hazard Class 8  
Packing Group II

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component             | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-----------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Methyltetrahydrofuran | X                                                   | -                                       | X    | X     | 202-507-4 | X    | X   | X     | -    | X    | X    | KE-33479 |

## National Regulations

## SECTION 16. OTHER INFORMATION

Prepared By  
Revision Date

Health, Safety and Environmental Department  
12-May-2024

## 2,4-Difluorobenzylmagnesium bromide, 0.25M in 2-Methyltetrahydrofuran

## Revision Summary

Not applicable.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Legend**CAS** - Chemical Abstracts Service**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**NZIoC** - New Zealand Inventory of Chemicals**WEL** - Workplace Exposure Limit**ACGIH** - American Conference of Governmental Industrial Hygienists**DNEL** - Derived No Effect Level**RPE** - Respiratory Protective Equipment**LC50** - Lethal Concentration 50%**NOEC** - No Observed Effect Concentration**PBT** - Persistent, Bioaccumulative, Toxic**TWA** - Time Weighted Average**IARC** - International Agency for Research on Cancer**PNEC** - Predicted No Effect Concentration**LD50** - Lethal Dose 50%**EC50** - Effective Concentration 50%**POW** - Partition coefficient Octanol:Water**vPvB** - very Persistent, very Bioaccumulative**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road**OECD** - Organisation for Economic Co-operation and Development**BCF** - Bioconcentration factor**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code**MARPOL** - International Convention for the Prevention of Pollution from Ships**ATE** - Acute Toxicity Estimate**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Physical hazards

On basis of test data

## Health Hazards

Calculation method

## Environmental hazards

Calculation method

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**